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Ultrabroadband: the next stage in communications

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Ultrabroadband: the next stage in communications

The Transition Toward Intensive Network Technologies: a Macroeconomic Perspective (*)

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Abstract: This article analyses the macroeconomic effects of the introduction of a new technology that exhibits strong network effects in a sectoral economy. We show that the endogenous adoption of such a technology can lead to cross-sector relocation of resources of the kind industrialized economies have experimented in the last 40 years. We also show that contrary to the common wisdom, such a technological change does not harm employment in the medium and the long term, and is welfare improving.

Key words: macroeconomic dynamics, technological change, network effects.

Ultrabroadband everywhere is an upcoming evolution of broadband which requires a significant amount of investment that can in turn magnify the efficiency of informational and procurement processes in the whole economy. However, the kind of incremental innovations over ultrabroadband technology that will unleash its potential benefits are not known yet. The deployment and further technological upgrading of a new generation of access network will certainly allow for a significant increase in the capacity of information storage and diffusion. But it is also commonly perceived that this process will require further investment in physical capital. Henceforth, there exists widespread agreement that a dynamic transition between old and new infrastructure devices should prove harmful to the level of employment while improving production efficiency. In order to assess the global impact of ultrabroadband on productivity and on employment, we propose to model the deployment of ultrabroadband as a generic transition toward a new technology, which features *(i)* potentially strong network

(*) We would like to thank the participants of the seminar on "UltraBroadBand: The next generation of infrastructure and applications", April 2008, Paris.

externalities, *(ii)* higher returns on physical capital and *(iii)* is subject to the random occurrence of incremental innovations. By doing so we aim at considering a technology which properties can easily be related to the characteristics of ultrabroadband.

This paper studies the effects of technology adoption on the economy. In particular, we are interested in the evolution of the main aggregates - output and unemployment - in the aftermath of the adoption of a new technology that is relatively more capital intensive than the former. [1] According to common wisdom, the adoption and the productive use of a more capital intensive technology usually generate larger unemployment. Our analysis aims at investigating whether this view is valid in the context of ICT technologies. We also aim at investigating whether the adoption of a more capital intensive technology that requires a greater effort in terms of accumulation leads to a situation where households fare necessarily worse off.

We build and estimate a dynamic general equilibrium model of the technological transition in the form of the adoption of a new general purpose technology. The new technology features two main characteristics that are usually associated with ICT technologies. First it is more capital intensive, implying that the marginal returns to capital accumulation are larger. Second the new technology is characterized by the existence of important specific external network effects: The larger the diffusion, the higher the productivity gains associated with the productive use of the new technology. We then estimate the model for the French economy over the last 25 years and show that it can account for the recent trends in the evolution of sector shares. We find that the adoption of the new capital intensive technology exerts a positive effect on the rate of output growth. Interestingly, we also find that positive network externalities dominate the potential congestion effects associated with technology diffusion. In other words, the sole specific external network effects embodied in the new technology can account for the long run positive economic impacts of technological renewal and further upgrading. In particular, such effects prevent the unemployment rate from increasing. Moreover, in the long run, such specific effects actually played in favor of higher employment in the long-run. The common wisdom usually perceives the adoption of a more capital intensive technology as detrimental for employment. Our results indicate that, on the contrary, the adoption of

[1] As is the case for most new ICT technologies.

such technology has positive effects on both the unemployment rate and the social welfare as well.

The first section presents the model. The second section details our calibration-estimation procedure to assign values to the parameters of the model. In particular we highlight how one can use aggregate information to reveal parameters pertaining to the new technology. The third section is devoted to the study of the aggregate dynamic implications of technological adoption and provides an analysis of the effects of an increase in the magnitude of network external effect related to the productive use of an ICT technology. A last section offers some concluding remarks.

■ The model

The model consists of a multi-sectors closed economy *à la* HUFFMAN (1993). In each and every period, sectoral firms have access to two technologies. The first one is the "old" production technology. It is a mature technology which productive efficiency does not improve over time. The second one is defined by a new, more capital intensive technology that firms may choose to adopt or not. This adoption decision is found to be the main driving force of the economic growth process. [2]

Households

The economy is populated by a continuum of households. Without any loss of generality, we assume that the size of population is kept constant and is normalized to one. At the beginning of each and every period, the agents face a probability ψ>0 of dying. According to the law of large number a share ψ of the population disappears at the end of each period. At the same time a share ψ is born so as population is kept constant. At the beginning of period t, the household is told whether she will survive to the next period, in which case, she consumes c_t of a final good, or die. In that case she consumes d_t. Assuming that the household's instantaneous preferences are represented by a logarithmic function, the expected utility of an individual over her total expected life time is given by:

[2] A technical appendix available from the authors details the model.

$$\mathcal{U}_{n,t} = \sum_{s=0}^{\infty} \rho^s (1-\psi)^s \log(c_{n,t+s}) + \sum_{s=1}^{\infty} \rho^s \psi (1-\psi)^{s-1} \log(d_{n,t+s})$$

where $\rho \in (0,1)$ is a constant discount factor.

When an agent is born, she is endowed with a fixed quantity of labor, $\bar{\ell} = 1/\varphi$, she supplies inelastically on the labor market in the very first period of her life only. She may or may not be selected for work, according to a matching process à la Pissarides [1990] that shall be described further. If selected on the job market, the householder works and receives the real wage, W_t. This income is then used to pay taxes, τ_t, to finance unemployment insurance, to consume, $c^e_{t,t}$ and to accumulate assets, $a^e_{t,t}$. [3] When unemployed, the agent receives unemployment spells, Ω_t. Such as the worker, the agent allocates her individual income between the payment of taxes, own private consumption, and assets accumulation. Therefore, the budget constraint of the agent in her very first period of life takes the form:

$$c^e_{t,t} + a^e_{t,t} = (W_t - \tau_t)\bar{l} \text{ when employed}$$
$$c^u_{t,t} + a^u_{t,t} = (\Omega_t - \tau_t)\bar{l} \text{ when unemployed}$$

In the remaining part of her life, the agent does not work anymore and acts as a capitalist. She then gets her revenues from interest rate payments, r_t. The budget constraint is then given by $c_{n,t} + a_{n,t} = (1+r_t)a_{n,t-1}$ when the agent survives to the next period, and $d_{n,t} = (1+r_t)a_{n,t-1}$ when the agent dies at the end of the period.

The household then determines her consumption/saving plans by maximizing her expected life time utility function subject to the budget constraints.

Employment decision

The level of employment is determined by a matching process *à la* MORTENSEN (1986) and PISSARIDES (1990). Trade in the labor market is costly and imperfectly coordinated across agents. To capture such market

3 In this notation, $x^i_{s,t}$ denotes the quantity x of an agent born in period s at time t when her status on the labor market is i (i=e when the agent is employed, i=u when the agent is unemployed).

failures, we assume that a costly search process exists on the labor market. More precisely, a concave matching function exists that relates the number of hirings, $\bar{L}_t$, on the labor market to the aggregate level of unemployment, $\bar{U}_t$, and the aggregate level of vacancies, $\bar{V}_t$, posted by each firm during the period,

$$\bar{L}_t = m_0 \bar{U}_t^{\eta} \bar{V}_t^{1-\eta} \text{ with } 0 < \eta < 1$$

At the individual level each firm i controls the level of employment by altering its vacancy posting policy, $V_{i,t}$. The firm, however, faces an externality related to the job market trading process: Due to coordination failures in the job posting policies of firms, no firm is able to control the probability, q_t, of filling a vacancy. Additionally, posting a vacancy triggers a positive cost, $\omega_t > 0$, per vacancy.

The final good sector

An infinite number of atomistic firms exist producing a homogenous final good, Y_t All firms are perfectly identical such that we will assume a representative firm exists that acts in a competitive manner on both the input and output markets. The final good is produced according to a constant return to scale technology combining the goods, $Q_{i,t}$, produced in the N sectors of the economy. This technology is represented by the CES production:

$$Y_t = \left(\sum_{i=1}^{N} \zeta_i^{1-\varepsilon} Q_{i,t}^{\varepsilon} \right)^{1/\varepsilon}$$

where $\varepsilon < 1$, $0 < \zeta_i < 1$. The parameter ε determines the degree of substitutability between the sectoral goods, and $\zeta_i$determines that actual share of intermediary good i in final output. The composition of the aggregate is then determined by the maximization of the representative firm's profit.

The final good can be used for consumption and investment purposes. Investment is used to form the capital stock according to the standard capital accumulation law:

$$K_{t+1} = I_t + (1 - \delta) K_t$$

where $\delta \in (0,1)$ is the depreciation rate.

Sectoral firms

The economy consists of N sectors producing a specific good in quantity $Q_{i,t}$. Each sector is populated by a large number of identical firms that act in a competitive manner both in the good market and in the input market. Without loss of generality, we assume there is a representative firm for each sector i. Each sectoral good, $Q_{i,t}$, is produced using capital, $K_{i,t}$ and labor, $L_{i,t}$, according to a constant returns to scale technology. First, firms have access to an existing leading technology represented by the Cobb--Douglas production function:

$$Q_{1,i,t} = K_{i,t}^{\alpha_i} L_{i,t}^{1-\alpha_i}$$

where $0 \leq \alpha_i \leq 1$.

A new technology then becomes available to the firms:

$$Q_{2,i,t} = \Phi_{i,t} K_{i,t}^{\beta_i} L_{i,t}^{1-\beta_i}$$

where $0 \leq \alpha_i \leq \beta_i \leq 1$ and $\Phi_{i,t} > 0$.

This formulation calls for two comments. First of all, it is implicitly assumed that technological progress takes the form of the diffusion of a relatively more capital intensive production technology, $\alpha_i \leq \beta_i$. In other words, the new technology is a capital intensive technology. Second, $\Phi_{i,t}$ denotes the relative total factor productivity of the new technology which dynamics takes the form:

$$\Phi_{i,t} = \gamma_{i,t}^{\varphi} \Phi_{i,t-1}$$

where $\gamma_{i,t}^{\varphi}$ - the growth factor of relative productivity - is given by:

$$\gamma_{i,t}^{\varphi} = \begin{cases} \bar{\gamma}_i^{\varphi} \text{ with probability } \pi_t \\ 0 \text{ with probability } 1 - \pi_t \end{cases}$$

This process is observationally equivalent to the occurrence of incremental innovations that enhance total factor productivity of the new technology. The probability, π_t, of an improvement of the new technology is endogenously determined and takes the form:

$$\pi_t = \rho_\pi \pi_{t-1} + \pi^*(\bar{\varsigma}_{k,t})$$

$\pi^*(\bar{\varsigma}_{k,t})$ is the endogenous part of the probability evolution, which is characterized by the existence of an external network effect. More precisely,

$\pi^*(.)$ is a function of the share of physical capital allocated to the new technology:

$$\pi^*(\bar{\varsigma}_{k,t}) = (1-\varpi_0)\bar{\varsigma}_{k,t} - \bar{\varsigma}_{k,t}{}^{1+\varpi_1} + \varpi_0 \text{ where } \varpi_0, \varpi_1 > 0$$

When all physical capital is allocated to the old technology, the network effect is nil. The probability is then totally exogenous and fixed to the constant ϖ_0. When all existing capital stock is allocated to the new technology, this component is nil. During the adoption process - as $\bar{\varsigma}_{k,t}$ increases - the probability first increases, therefore the higher the adoption rate of the new capital intensive technology, the higher the probability of productivity improvements related to the use of the new technology. After a given threshold, $\bar{\varsigma}_k = \left(\frac{1-\varpi_0}{1+\varpi_1}\right)^{1/\varpi_1}$, positive external network effect turns into congestion effect and $\pi^*(.)$ is a decreasing function of the adoption rate.

Three cases can then arise:

- Only the old technology is implemented, and the production function is given by:

$$Q_{i,t} = Q_{1,i,t} = K_{i,t}^{\alpha_i} L_{i,t}^{1-\alpha_i}$$

- Only the new technology is implemented. The production function is then:

$$Q_{i,t} = Q_{2,i,t} = \Phi_{i,t} K_{i,t}^{\beta_i} L_{i,t}^{1-\beta_i}$$

- The two technologies are implemented in each sector. A share $\sigma_{i,t}$ (resp. $\upsilon_{i,t}$) of sector i physical capital (resp. labor) is allocated to the old technology, while the remaining share is being allocated to the new technology:

$$Q_{i,t} = (\sigma_{i,t} K_{i,t})^{\alpha_i} (\upsilon_{i,t} L_{i,t})^{1-\alpha_i} + \Phi_{i,t}((1-\sigma_{i,t})K_{i,t})^{\beta_i}((1-\upsilon_{i,t})L_{i,t})^{1-\beta_i}$$

The sectoral firm then decides on its production/technology plans by maximizing its profit. The technology decision is then characterized by the following proposition. [4]

Proposition 1: There are two levels of capital/labor ratios, $\underline{k}(\Phi_{i,t})$ and $\bar{k}(\Phi_{i,t})$, such that:

[4] The proof is available from the authors upon request.

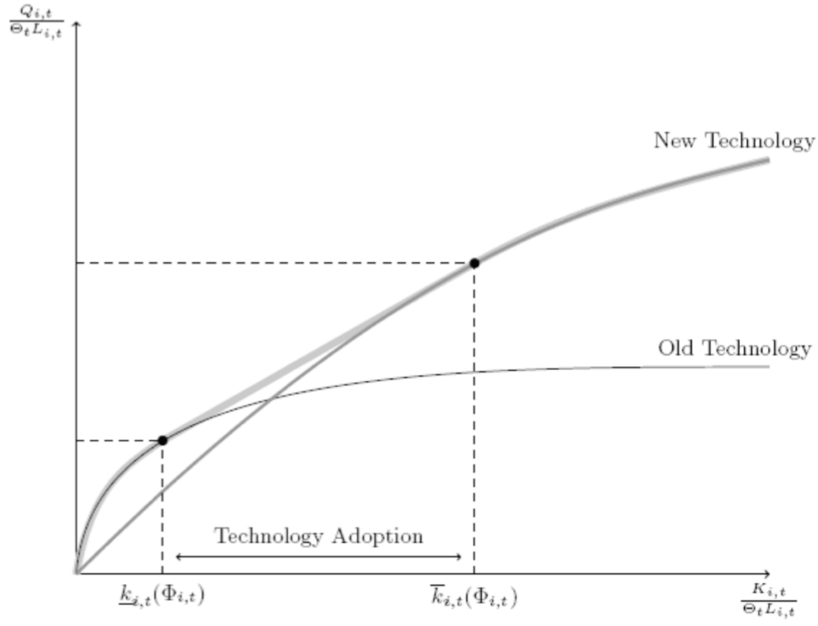
- if $\frac{K_{i,t}}{L_{i,t}} < \underline{k}(\Phi_{i,t})$, only the old technology is implemented
- if $\frac{K_{i,t}}{L_{i,t}} > \overline{k}(\Phi_{i,t})$, only the new technology is implemented
- if $\underline{k}(\Phi_{i,t}) \leq \frac{K_{i,t}}{L_{i,t}} \leq \overline{k}(\Phi_{i,t})$, the two technologies are used

where $\underline{k}(\Phi_{i,t}) = \Phi_{i,t}^{\frac{1}{\alpha_i-\beta_i}}\left(\frac{\alpha_i}{\beta_i}\right)^{\frac{\beta_i}{\alpha_i-\beta_i}}\left(\frac{1-\alpha_i}{1-\beta_i}\right)^{\frac{1-\beta_i}{\alpha_i-\beta_i}}$

and $\overline{k}(\Phi_{i,t}) = \Phi_{i,t}^{\frac{1}{\alpha_i-\beta_i}}\left(\frac{\alpha_i}{\beta_i}\right)^{\frac{\alpha_i}{\alpha_i-\beta_i}}\left(\frac{1-\alpha_i}{1-\beta_i}\right)^{\frac{1-\alpha_i}{\alpha_i-\beta_i}}$

This proposition is illustrated in figure 1 and is critical for the rest of our analysis as it determines the conditions under which a firm decides to adopt the new technology.

Figure 1 - Technology decision



It clearly states that the decision related to technological adoption depends on several specific factors. First, it depends on the the level of the relative total factor productivity of the new technology in sector i, $\Phi_{i,t}$. More precisely, the more productive the new technology, the easier it is for the firm to adopt it. In other terms the more mature the new technology (the larger $\Phi_{i,t}$), the lower the capital/labor ratio required to adopt it. Interestingly, the level of the relative productivity exerts a symmetrical effect on the upper and the lower thresholds. In other words, a shift in $\Phi_{i,t}$ just leads to a translation of the adoption zone without widening or narrowing it. It therefore affects the starting period of adoption, but leaves the length of the adoption process unaffected. Second, the accumulation process plays a key role in the adoption decision as the latter is essentially determined by the capital/labor ratio. All things being equal, further accumulation effort makes the new technology adoption more profitable to the firm. Similarly, the existence of a high unemployment rate eases adoption, which suggests that there may be a trade-off between capital intensive technology adoption and employment. Finally, the relative prices of capital and labor are critical to the understanding of the adoption process. For instance, a higher wage - by discouraging employment - makes technology adoption more beneficial to the firm.

The linearity of the production function during the adoption phase (see figure 1) enlightens the fact that the allocation of capital and labor is kept optimal by the firm, allowing to escape from the marginal decreasing returns zone.

General equilibrium

A general equilibrium of this economy is a sequence of prices $\{\mathcal{P}_t\}_{t=0}^{\infty} = \{r_t, W_t, p_{k,t}, \omega_t\}_{t=0}^{\infty}$ and a sequence of quantities $\{\mathcal{Q}_t\}_{t=0}^{\infty} = \{c_t, d_t, a_t, C_t, A_t, I_t, K_{t+1}, Y_t, Q_{i,t}, K_{i,t}, L_{i,t}, \sigma_{i,t}, v_{i,t}; i = 1, \dots, N\}_{t=0}^{\infty}$ such that:

- given a sequence of prices $\{\mathcal{P}_t\}_{t=0}^{\infty}$, the sequence of quantities solve the household and the firms problems;
- given a sequence of quantities $\{\mathcal{Q}_t\}_{t=0}^{\infty}$, the sequence of prices clear the markets;
- the labor market satisfies the restrictions implied by the matching process.

Calibration and estimation of the model

The quantitative analysis of the model requires assigning numbers to the deep parameters. Some parameters - those pertaining to the household, the labor market and the production process prior to 1979 - are calibrated. All parameters related to the new technology are estimated relying on a simulated non-linear least squares technique. Both the calibration and the estimation of the model are undertaken using annual French data for the time period 1979-2003. The data are borrowed from the "60-Industry Database" published by Groningen Growth and Development Centre. [5] We consider the three main sectors of the economy: Agriculture, Industry and Services.

Calibrated parameters

The death probability of an agent, ψ, is set so as to guarantee an agent's average economic lifetime of 50 years. Since this average lifetime is given by $1/\psi$, ψ is set to 0.02. The psychological discount factor, ρ, is set such that the model correctly reproduces the average aggregate capital/output ratio of the French economy during the period 1978-2003. This ratio is 2.795 which implies ρ=0.8489.

The parameters related to the matching technology and the wage setting process are set so as to match labor market facts. The elasticity, η, of the matching function is borrowed from FÈVE & LANGOT (1996) who estimate a general equilibrium model of the French economy. This leads us to set $\eta = 0.4$. The scale parameter of the matching function, m_0, is set so as to match the unemployment rate in 1979 (6%), leading to $m_0 = 59.5556$. Unemployment spells, Ω_t, are assumed to be a linear function of the aggregate wage ($\Omega_t = \tau W_t$). Following Wasmer [1999], we assume that the hiring costs faced by the firms account for 3.3% of the total labor compensation. Using this information we set the replacement ratio τ to 0.978. This may appear way too high compared to compensation ratios reported by the OECD. It must however be kept in mind that, in this model, Ω_t actually captures all external opportunities an agent may face on the labor market (in particular home production, self-employment opportunities, …). Finally the unit cost of a vacancy faced by a firm is set such that the

[5] Data can be freely downloaded from http://www.ggdc.net/index-dseries.html.

model matches the French vacancy to unemployment ratio over the period (0.108). This implies a unit vacancy cost (ω) of 3.2492.

The production of the final good requires setting a value to N+1 parameters : ε and ζ_i, i=1, …, N. In our case, we have N=3. ε is set to 0.5. [6] The share of each sectoral good in the final good aggregate, ζ_i, is set such that in the steady state, the model reproduces the share of each sector in total GDP of the French economy in 1979. This led us to set ζ_i to 0.0157, 0.2957 and 0.4926 for, respectively, Agriculture, Industry and Services.

The parameter, α_i, determining capital intensity in the old technology for each sector is set such that - assuming only the old technology is available to the firms in 1979 - the model reproduces the 1979 French sectoral labor shares in the steady state. This led us to set α_i to 0.8386, 0.2376 and 0.3864 in , respectively, Agriculture, Industry and Services.

Finally, the depreciation rate δ is set such that physical capital depreciates at an annual rate of 6%, which corresponds to an average life of capital of about 16.66 years. [7]

Estimation procedure

The parameters characterizing the new technology are estimated using a simulation technique. Indeed, *i)* the exact form of the new technology is not known (at least at the aggregate level) and *ii)* most of the parameters involved in the description of the new technology do not have any directly observable counterpart. [8]

We therefore estimate the sector capital elasticities of the new technology, β_i, and the parameters related to the relative productivity shocks, $\varpi_0, \varpi_1, \rho_\pi, \gamma_i^\varphi$. We also estimate the initial level of the relative total factor productivity in each sector. More precisely we estimate the, Δ,

[6] Simulation exercises using alternative values of ε showed that the results are robust to changes in the values of this parameter.

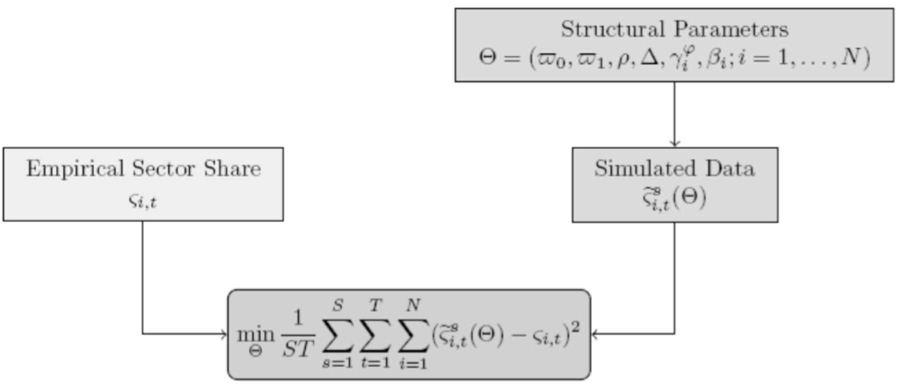
[7] It should be also noted that the model is characterized by the existence of capital and labor adjustment costs for purely technical reasons. They are however set to a tiny level, and we therefore do not discuss their exact form to save on notations and space. A more detailed exposition of the model is however available from the authors upon request.

[8] This is in particular the case for the parameters related to the probability of occurrence of a productivity shock and the size of network externalities.

distance between the initial level and the actual level of the relative productivity that makes the firm indifferent between adopting the new technology and still relying on the former.

The vector of parameters, $\Theta = (\varpi_0, \varpi_1, \rho_\pi, \Delta, \gamma_i^\varphi, \beta_i; i = 1, \dots, N)$, is estimated by a simulated non--linear least square technique which is described in figure 2. The purpose of this method is to determine the set of structural parameters that minimize the discrepancy between actual and fitted data, the latter being derived from model's simulation.

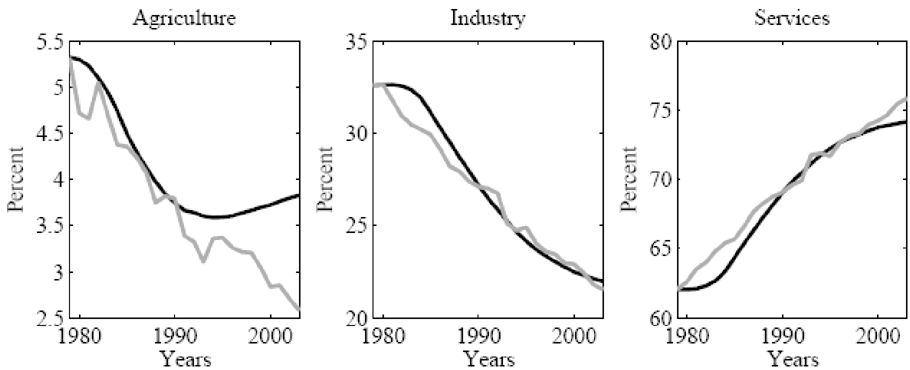
Figure 2 - Estimation algorithm



Estimation results

Figure 3 compares the dynamics of sectoral shares as predicted by the model (plain dark line) to the actual data (plain grey line). As can be seen from the graph the model provides a relevant framework to account for the overall dynamics of sectoral shares over the last 25 years. In particular, the model accounts for the the observed decline in the respective shares of Agriculture and Industry, and remarkably mimics the upsurge of Services over the last years. In other words, the emergence of a new capital intensive technology that incorporates external network effects accounts for the increasing diffusion of sector of services in the French economy.

Figure 3 - Sectoral shares



Note: Plain dark line: model estimates. Plain grey line: French data.

Tables 1 and 2 report estimates of parameters associated with the new technology. A first implication derived from these estimates is that the new technology was not mature in 1979, as its relative productivity lay about 9% below the level that would have made it worth adopting (Δ =0.91). Table 1 also reveals that the dynamic process of the probability of occurrence of incremental innovations exhibits persistence (ρ_π =0.8). In particular, the mean lag of these dynamics - absent of any network effect - is about 5 years. The probability of productivity improvement is initially relatively high, about 12% (ϖ_0=0.117).

Table 1 - Productivity process

ρ_π	ϖ_0	ϖ_1	Δ
0.7991	0.1168	0.5761	0.9113

It is important to observe that the technology exhibits strong network effects, as the elasticity of the probability process to aggregate adoption is about 0.6 (ϖ_1=0.5761). Therefore, when the speed of the adoption process attains its peak, the network effect drives the probability of productivity improvement up to 60%. Interestingly, if we shut off the network effects, the model becomes totally unable to account for observed evolution of sectoral shares. In other words, the new technology investigated in this framework exhibits significant network effects associated with the productive use and the diffusion of ICT technologies.

Table 2 reports estimated values for the magnitude of productivity improvements generated by incremental innovations, γ_i^φ, and the marginal returns to capital in the new technology, β_i, in each sector. First of all, this table clearly shows that average productivity gains are larger in Services sector than in both Agricultural and Industrial sectors (10.8% vs. 1.8% and 3.3%) These values are in accordance with the common wisdom that ICT technologies first generated larger productivity gains in Services.

Table 2 - Sectoral parameters

	Agriculture	*Industry*	*Services*
γ_i^{φ}	0.0180	0.0334	0.1080
β_i	0.9222	0.3046	0.4977

The estimated values for the marginal returns of capital, β_i indicate that the new technology is more capital intensive than the old technology. Note that this result is independent from the estimation constraints and therefore confirms the higher capital intensity of the new technology with respect to the former. It is also worth noting that this increase in the role of capital accumulation displays a significant degree of heterogeneity across sectors. For instance, while the returns to capital have increased by 10% in Agriculture, they experienced a 28.16% increase in Industry and 28.81% in Services, which is the sector that benefits the most from the new technological paradigm.

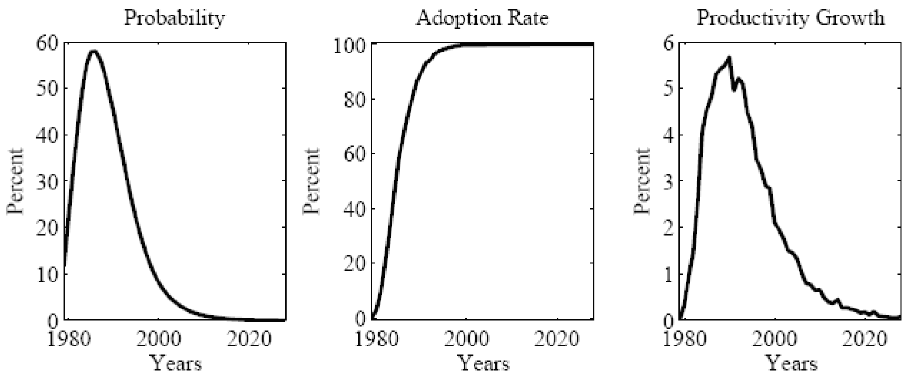
■ Macroeconomic implications of technology adoption

The preceding section established that the model provides a relevant framework allowing the understanding of the recent evolution of the French economy in light of technological upgradings. In this section, we investigate the macroeconomic implications of the technological transition towards the new technology on the growth process and the labor market.

A benchmark simulation

Figure 4 reports the evolution of the probability of occurrence of incremental innovations, aggregate adoption and output growth as predicted by the model. It is important to stress that since the model only features a technological change, leaving aside any policy issues or institutional changes, the dynamics, as generated by the model, provide information about the very effect of technological change on the main macroeconomic variables absent from any other change.

Figure 4 - Aggregate dynamics

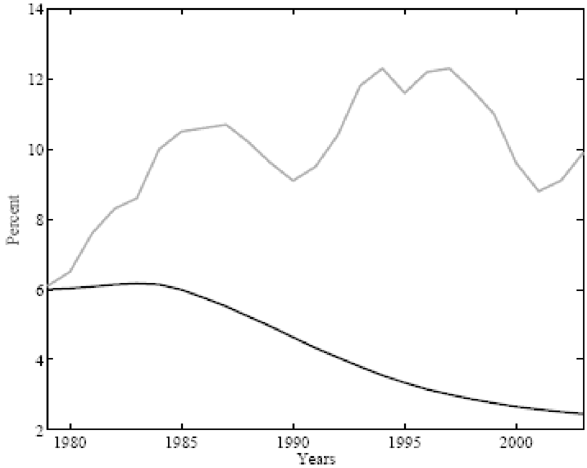


Note: Plain dark line: model estimates. Plain grey line: French data.

Let us first assume that the economy is initially in a steady state where only the old technology is in use. Because $\varpi_0 > 0$, the relative productivity of the new technology, $\Phi_{i,t}$, increases in all sectors. Eventually, the level of productivity related to the productive use of the new capital intensive technology reaches a level which appears to be high enough to make it profitable for one sector to start the new technology adoption process. This occurs first in Services, as this is the sector where marginal productivity improvements, γ_i^{φ}, are the largest. Services therefore allocate a greater share of available capital to the productive use of the new technology, so that the aggregate share of capital allocated to the new technology increases, hence the endogenous part of the probability. In turn, the emergence of network effects enhances the probability of increases in $\Phi_{i,t}$ in all sectors (See left panel of figure 4). In other words, the average growth rate of total factor productivity associated with the new technology increases and $\Phi_{i,t}$ accelerates. As shown in Proposition 1, this induces a shift of the adoption zone to the left. Firms are then more likely to adopt the new technology. The Industry sector then starts adopting, followed by Agriculture. This reinforces network effects, and the probability further rises. Adoption, as measured by the share of firms adopting the new technology, steadily increases (see central panel of figure 4). This has a clear implication in terms of output growth: during the adoption phase, as more and more firms adopt the new technology - the more productive technology - the more the growth rate of aggregate output increases. This occurs for two main reasons. First of all, as the probability of occurrence of incremental innovations increases - due to specific network effects - total factor productivity growth rises, hence output growth. Second, since the new technology is more capital intensive, the marginal returns to capital accumulation increase. As a result, it is worthwhile for the firms to accumulate physical capital. All things being equal, the capital/labor ratio does increase, raising the incentives to adopt the new ICT technology. At the peak, the very effect of technological adoption is to raise output growth up to 6% per year. [9]

Figure 5 reports the actual evolution of unemployment (plain grey line) as well as the evolution of unemployment as determined by the sole effect of the technological adoption process (plain dark line). Such a comparison is of great importance, as the common wisdom usually indicates that the adoption of a more capital intensive technology should lead firms to substitute capital for labor. In other words, a more capital intensive technology adoption is usually perceived as a job killing device.

Figure 5 - Unemployment dynamics



Note: Plain dark line: model estimates. Plain grey line: French data.

Figure 5 indicates that the very effect of adoption of the new technology was to lower the unemployment rate in the long run. At a first glance, this may appear as a paradoxical feature of the model. This is actually not the case as indicated by the very short-term dynamics. Indeed, in the very short-term firms do actually substitute capital for labor. As a result, the

[9] Again let us recall that this occurs absent from any other changes in the economy. Actual data indicate lower output growth, which actually reflects that other shocks or decisions affected the growth process.

unemployment rate increases slightly during the first five years of the adoption process. Furthermore, the adoption of the new technology appears to have another important effect: The increase in the returns to capital accumulation does increase the marginal productivity of labor. Therefore, it is worthwhile for the firms to increase their labor demand. In other words, the productivity effect counters the substitution effect. Due to the existence of positive network externalities, the productivity effect eventually dominates. Therefore, contrary to widespread wisdom, the adoption of a more capital intensive technology leads to a decrease in the unemployment rate. In others, the rise of French unemployment should not be attributed to the adoption of the new technology but rather to bad policies, institutional changes or bad shocks that hit the French economy.

An advantage of structural modeling is that it gives a clear measure of the potential gains/losses of technology adoption: welfare, as measured by the discounted sum of expected utility of agents (See eq. 1). In order to obtain an economic interpretation of the welfare gains, we express them in terms of consumption level. More precisely we compute the extra consumption a household should be given in a world without technological transition to guarantee her the same level of utility as in an economy that experienced technology adoption. [10] Such a measure is of great importance as there are many reasons to think that households can be worse off during transitions.

First of all, as indicated in figure 5, unemployment increase in the very short–term which proves harmful to the consumption level. Second, technology adoption triggers greater capital accumulation, which is financed by savings. This leads households to postpone consumption and therefore has a direct negative effect on welfare in the very short–run. Our computations indicate that individuals actually strongly benefit from technological adoption. More precisely, an individual living in an economy where the representative firms only rely on the former technology would benefit from a weaker welfare level than an individual living in the economy where the representative firm actually adopts the new capital intensive ICT technology. As a result, to achieve the same welfare level as the adopting framework provides, one should be given a permanent 24.66% increase in consumption, to compound the absence of productivity gains related to the

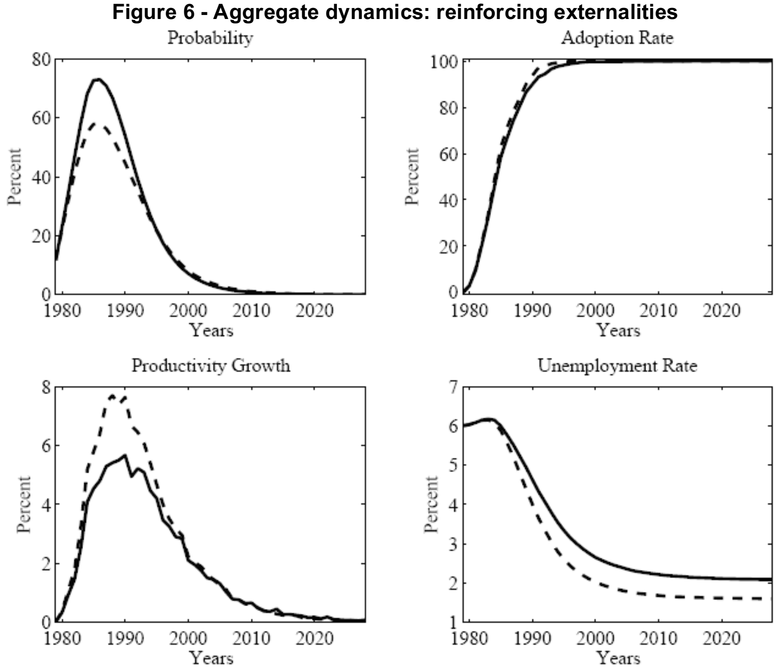
[10] From a technical point of view, this amounts to compute a transfer rate x such that $\mathcal{U}_t(c_t) = \mathcal{U}_t((1+x)c_t^*)$, where c_t (resp. c_t^* denotes the consumption path in the economy that experienced (resp. did not experience) a technology transition.

adoption process by the firm. Such high welfare level is directly related to the magnitude of the productivity gains triggered by the network external effects embodied in the new ICT technology.

Increasing network externalities

In this section, we consider an increase in the size and magnitude of network effects related to the deployment, diffusion and use of a new generation of electronic communication technologies, such as broadband and ultrabroadband wire line and wireless network access. In particular, we increase the size of the elasticity of $\pi^*(.)$ to the aggregate share of capital allocated to the new technology, ϖ_1, by a factor 2.

Figure 6 - Aggregate dynamics: reinforcing externalities



Note: Plain dark line: Benchmark experiment. Plain grey line: Higher external network effect.

Figure 6 compares the evolution of the probability of occurrence of an incremental innovation, technological adoption, output growth and unemployment in our benchmark experiment (dashed line) and in the case of a larger network externality (plain line). The evolution of the probability provides a measure of the importance of the external network effect: by doubling ϖ_1 we increase the peak effect of adoption of the probability by 25%. More important, the increase in the magnitude of the external network effect does not have a significant effect on the timing of accumulation as shown in the upper right panel of the figure. The increase in the order of magnitude of such specific network effects has only a small impact on both the timing and the shape of the adoption process evolution. On the contrary, output growth is magnified. Indeed, although the speed of the adoption process is only marginally affected, the increase in the network externality has a large effect on the probability of occurrence of a technology shock. This raises the average growth rate of total factor productivity in the new technology, hence output growth. It is worth noting that the increase in output growth is much larger than the increase in the probability. At the peak of its time evolution, the growth rate of the economy is higher in the economy which technology features external network effects, with respect to the outcome of the benchmark simulation. The observed differential is 40%. . This result clearly shows evidence of a specific significant contribution of the network external effects embodied in the capital intensive ICT technology. Hence, due to larger productivity effects, the adoption of a new more capital intensive ICT technology, featuring larger external network effects such as, for example, broadband access network wire line and wireless technologies, may not have harmful effects on the aggregate economy.

■ Conclusions

This paper shows that the adoption of a highly capital intensive technology that exhibits significant external network effects has significant and positive impacts on the aggregate economy in the long run. Such properties are sufficient for the firms to generate high enough productivity gains [11] that drive the unemployment rate downward in the long run. Most important, the formal results derived from both the theoretical framework and simulations outcome show evidence of a highly significant impact of the adoption and productive use of the new technology on the aggregate economy. Such high magnitude order effects are directly related to this remarkable property of the new available ICT technology: the embodied relatively larger positive external network effects are responsible for the size and magnitude order, as well as for the speed and the depth of productivity

[11] At the middle of the sample period, productivity gains prove to have increased by 6%.

gains diffusion across all sectors of the aggregate economy. In this case, the higher the positive network effect, the larger the productivity gains. As the adoption process requires a larger amount of capital, the capital deepening increases with the rate of adoption. As a result, this capital/labor substitution process has a negative effect on employment in the short run. In the long run, though, the increasing size and magnitude of network effects raises the probability of an incremental innovation. This increases the expected productive efficiency of the new technology. As a result, since the new technology exhibits higher returns on capital and accumulation increases, so does the marginal return to labor. Firms post more vacancies and therefore the unemployment rate eventually declines. Furthermore, at the private household level, such productivity gains allow to compound the cost of the trade off between current consumption and the financing of capital accumulation.

According to the results, the adoption of a new ICT technology has strong and significant long run positive effects on both the efficiency of the production sector and the rate of unemployment, and on the aggregate level of welfare as well. These results provide a deeper understanding of the dynamic links between the diffusion of a communication access network technology and the resulting overall efficiency gains. Moreover, these effects are able to account for the historical evolution of the French economy. Further research should aim at capturing other specific properties of ultrabroadband capital, in order to isolate its very contribution on growth, efficiency gains and welfare. In particular, this may reveal the specific effects of digitalization on the job matching process (skills, training, learning processes, …).

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